

Alpha Diversity

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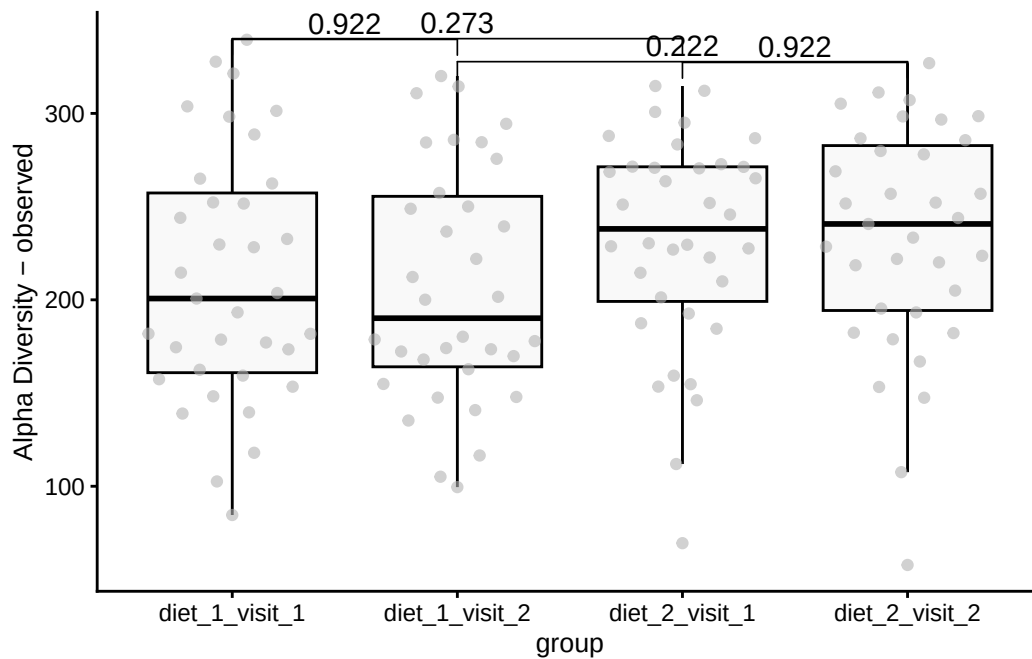
Introduction

This document reports on a microbiota project results of alpha diversity across different diets and visits.

Group-wise comparisons

- **Index:** observed
- **Group variable:** group
- **Multiple testing correction:** fdr

Visualization



Characteristic	after			before		
	Beta	95% CI ¹	p-value	Beta	95% CI ¹	p-value
(Intercept)	144	59, 229	0.001	152	67, 236	<0.001
meal_group			0.3			0.4
1	—	—		—	—	
2	28	-16, 72	0.2	32	-11, 75	0.14
3	30	-14, 73	0.2	27	-15, 70	0.2
4	-3.8	-47, 39	0.9	10	-32, 53	0.6
age	1.2	-0.25, 2.7	0.10	1.0	-0.45, 2.5	0.2

¹CI = Confidence Interval

Paired test of before and after	Beta	95% CI ¹	p-value
(Intercept)	147	89, 206	<0.001
meal_group			0.11
1	—	—	
2	28	-2.7, 58	0.074
3	28	-2.5, 58	0.072
4	2.2	-28, 32	0.9
age	1.2	0.13, 2.2	0.026

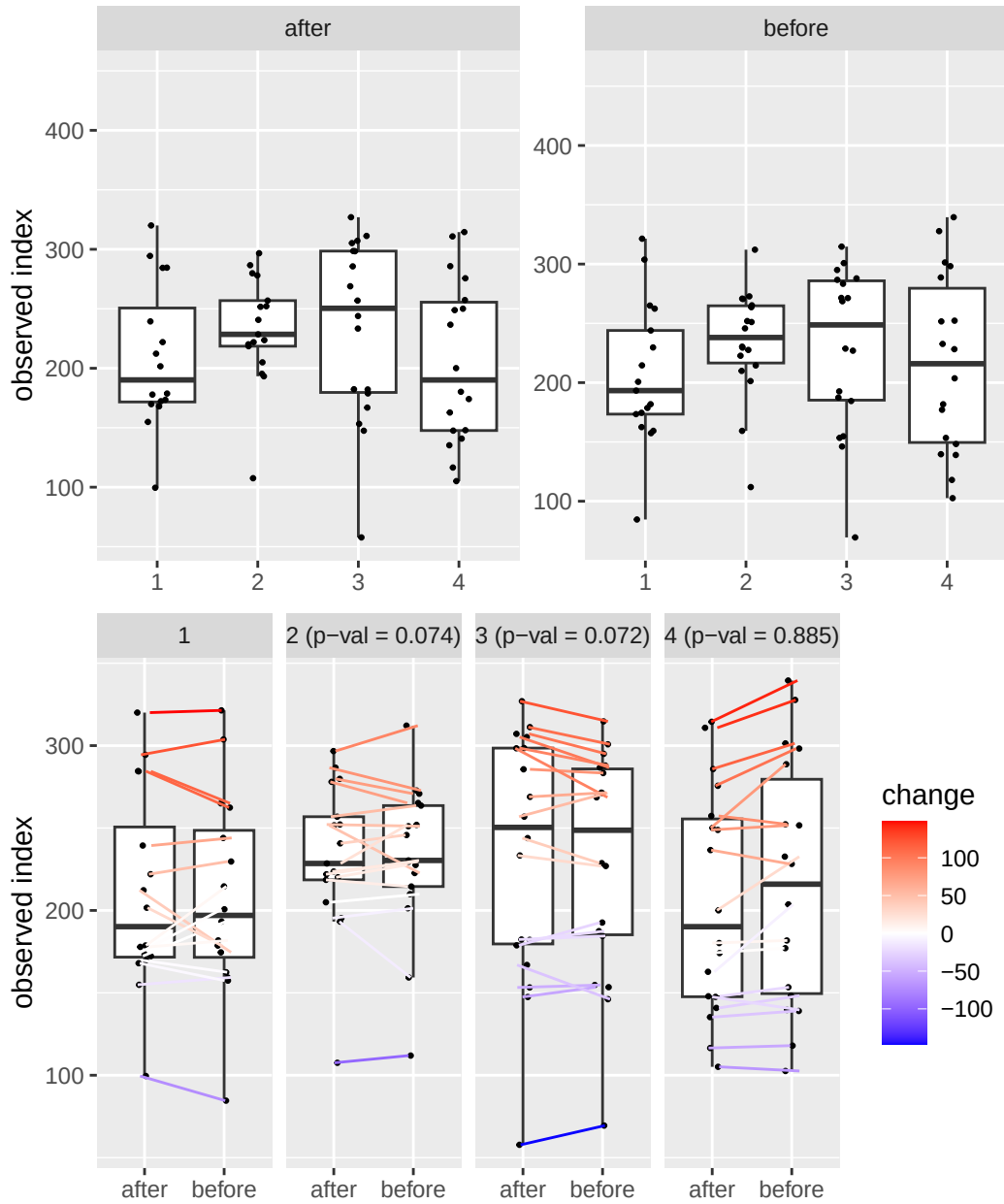
¹CI = Confidence Interval

Table of the top findings

Table 1: Table of alpha diversity comparisons

group1	group2	mean1	mean2	logFC	p.adj
diet_1_visit_1	diet_1_visit_2	211.16	207.15	-0.03	0.922
diet_2_visit_1	diet_2_visit_2	231.52	233.18	0.01	0.922
diet_1_visit_1	diet_2_visit_1	211.16	231.52	0.13	0.273
diet_1_visit_2	diet_2_visit_2	207.15	233.18	0.17	0.222

Alternate analyses



Data Summary

class: TreeSummarizedExperiment
dim: 1700 140

```

metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(13): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(10): PrevalentGenus PrevalentSpecies ... species strain
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL

```

Version info

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R version 4.4.2 (2024-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.1 LTS

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Matrix products: default
BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0

```

```

locale:
 [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
 [9] LC_ADDRESS=C             LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

```

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time zone: Europe/Helsinki
tzcode source: system (glibc)

```

```

attached base packages:
[1] grid      stats4    stats      graphics  grDevices  utils      datasets
[8] methods   base

```

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other attached packages:
[1] kableExtra_1.4.0      vegan_2.6-8
[3] lattice_0.22-6        permute_0.9-7

```

[5] lubridate_1.9.3	forcats_1.0.0
[7] stringr_1.5.1	purrr_1.0.2
[9] readr_2.1.5	tibble_3.2.1
[11] tidyverse_2.0.0	tidyr_1.3.1
[13] sechm_1.13.1	ComplexHeatmap_2.21.1
[15] scater_1.33.4	scuttle_1.15.4
[17] quarto_1.4.4	patchwork_1.3.0
[19] parameters_0.24.0	multtest_2.61.0
[21] miaTime_0.1.21	miaViz_1.15.4
[23] ggraph_2.2.1	mia_1.15.6
[25] TreeSummarizedExperiment_2.13.0	Biostrings_2.73.2
[27] XVector_0.45.0	SingleCellExperiment_1.27.2
[29] MultiAssayExperiment_1.31.5	SummarizedExperiment_1.35.3
[31] Biobase_2.65.1	GenomicRanges_1.57.1
[33] GenomeInfoDb_1.41.2	IRanges_2.39.2
[35] S4Vectors_0.43.2	BiocGenerics_0.51.3
[37] MatrixGenerics_1.17.0	matrixStats_1.4.1
[39] gtsummary_2.0.4	ggsignif_0.6.4
[41] ggpubr_0.6.0	ggplot2_3.5.1
[43] DT_0.33	dplyr_1.1.4
[45] cardx_0.2.2	ANCOMBC_2.8.0

loaded via a namespace (and not attached):

[1] randomcoloR_1.1.0.1	gld_2.6.6
[3] nnet_7.3-19	TH.data_1.1-2
[5] vctrs_0.6.5	energy_1.7-12
[7] digest_0.6.37	png_0.1-8
[9] shape_1.4.6.1	proxy_0.4-27
[11] Exact_3.3	registry_0.5-1
[13] rbiom_1.0.3	ggrepel_0.9.6
[15] bayestestR_0.15.0	parallelly_1.38.0
[17] mediation_4.5.0	MASS_7.3-61
[19] reshape2_1.4.4	foreach_1.5.2
[21] withr_3.0.2	xfun_0.49
[23] ggfun_0.1.6	survival_3.7-0
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[27] memoise_2.0.1	ggbeeswarm_0.7.2
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[31] gmp_0.7-5	tidytrees_0.4.6
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[35] gtools_3.9.5	V8_5.0.1
[37] Formula_1.2-5	datawizard_0.13.0
[39] httr_1.4.7	rstatix_0.7.2

[41] globals_0.16.3	ps_1.8.0
[43] rstudioapi_0.17.1	UCSC.utils_1.1.0
[45] generics_0.1.3	base64enc_0.1-3
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[49] zlibbioc_1.51.1	ScaledMatrix_1.13.0
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[97] lmom_3.2	GetoptLong_1.0.5
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[137]	coda_0.19-4.1	insight_1.0.0
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[141]	farver_2.1.2	tidygraph_1.3.1
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[171]	broom.helpers_1.17.0	treeio_1.29.1
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[175]	igraph_2.1.1	R6_2.5.1
[177]	broom.mixed_0.2.9.6	labeling_0.4.3
[179]	Rmpfr_1.0-0	cluster_2.1.6
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